Day 01 PySpark Interview Handbook

Topics Covered

Spark Ecosystem Hadoop vs Spark Spark Setup Introduction to Spark Introduction to PySpark

Q1. What is Apache Spark?

Interview Answer: Apache Spark is an open-source distributed data processing engine designed for fast, scalable analytics. It processes data across multiple machines in parallel and primarily performs computation in memory.

Real-world Example: A bank processing hundreds of millions of daily credit-card transactions uses Spark to partition data across executors, reducing processing time from hours to minutes.

Q2. Hadoop vs Spark

Interview Answer: Hadoop MapReduce is disk-based and optimized for batch processing. Spark performs most computations in memory, supports SQL, Streaming, ML, and Graph processing, and is significantly faster for iterative workloads.

Q3. Explain Spark Ecosystem

Spark Core, Spark SQL, Structured Streaming, MLlib, and GraphX together form the Spark ecosystem.

Q4. Explain Spark Architecture

Driver → Cluster Manager → Executors → Tasks. The Driver coordinates work, Executors execute tasks on partitions.

Q5. What is PySpark?

PySpark is the Python API for Apache Spark, allowing Python developers to build distributed data-processing applications.

This PDF is a starter handbook. A full version with 100+ interview questions, detailed answers, diagrams, production scenarios, and code examples will span 100+ pages and should be generated in multiple parts.